

WORD-ORDER VARIABILITY AND FSP IN WRITTEN AND SPOKEN DISCOURSE

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1. Although spoken and written language have long been recognised in modern linguistics as two partially autonomous subsystems of the general language system, rather than just a graphical and an acoustic representation respectively of essentially the same phenomenon, the descriptions of the differences between the two have been far from exhaustive. In this respect, *Longman Grammar of Spoken and Written English* (Biber et al., 1999, LGSWE henceforth) represents a notable exception to the general trend, using qualitative as well as quantitative analysis of an extensive corpus database and, as is clear from the title, devoting a separate section of the grammar to the description of the distinctive features of spoken language, namely Chapter 14: The Grammar of Conversation. However, attention to spoken language is not restricted to this chapter; references to grammatical features of conversation compared to those of fiction, news writing and academic prose are made throughout the book, typically complemented with frequency data of selected phenomena.

2. The characteristic features of spoken language stem from the fact that, unlike written language, the former is, as a rule, produced and perceived in real time, under time constraints and under constraints imposed by the span of short-term memory. It is interactive, constantly responding to the changing needs of ongoing communication and driven by the need to keep the communication flowing. As a result, spoken language is considerably less planned and contains a proportion of imperfect structures some of which are subsequently repaired.

Although spoken and written language, respectively, are often understood as homogeneous, it must be constantly borne in mind that both of these modes are internally varied and that different instances of either show the above characteristics in varying degrees (e.g. a piece of academic writing compared to a text message in writing, or a public speech compared to an informal conversation in speaking). Spokenness and writtenness must therefore be treated as graded phenomena.

The difficulty in analysing spoken language starts with the identification of the basic unit of analysis. In written language it has traditionally been the sentence, which, in addition to structural completeness, has other unambiguous boundary signals, such as gaps, punctuation, capitals, etc. In spoken language, the signals

are largely prosodic (tone-unit boundaries, pauses) and therefore not necessarily as distinct or absolute as their graphical counterparts. The LGSWE therefore uses the terms clausal and non-clausal units, rather than sentences, as the fundamental units of analysis in spoken language (Biber et al., 1999, 1039).

3.1. The different nature of spoken and written discourse is reflected in structural preferences, i.e. in the relative frequency of particular structures in speaking and writing, respectively. The differences are especially striking when conversation is compared with a quintessentially written register, academic prose. According to the LGSWE, the former abounds in the existential *there is* construction, the *wh-clefts*, and the *demonstrative wh-clefts* (e.g. *That's what I thought.*), whereas academic prose shows a markedly higher proportion of the passive and *it-clefts* (Biber et al., 1999, 938, 961, 963). The preference for certain syntactic choices in writing is probably motivated, among other factors, by the need to achieve semantic transparency, as well as by an effort to indicate unambiguously the information focus of the sentence.

3.2. Differences are also perceived in the choice of a particular linear arrangement of constituents within a given syntactic structure, i.e. word-order. The LGSWE observes that academic prose contains a significantly higher proportion of marked word-order than conversation (Biber et al., 1999, 963). This may be attributed to the role of word-order in indicating the information structure of the sentence, in which case the word-order has an effect similar to that of an informationally transparent syntactic structure, as well as to the need to achieve cohesion. In spoken language, the lack of informational transparency of syntactic structures is compensated by prosodic indicators, such as intonation (Firbas, 1992, 148).

4. The choice of both the syntactic structures and word-order options may serve the purpose of achieving a particular linear distribution of semantic units. This ordering is affected by two general principles in English: the principle of *end-focus* (Quirk et al., 1985, 1365), and the principle of *end-weight* (Quirk et al., 1985, 1282; see also Huddleston and Pullum, 2002, 1382–1383). These principles are not restricted to English. It has been demonstrated, e.g. by Dušková (1999, 281–288), that the principle of *end-focus* asserts itself even more strongly in Czech, owing to the greater mobility of clause constituents in the sentence. In English, the movement of a given semantic element into the final or near-final position is often impossible by simply rearranging the clause constituents within the same syntactic structure, but requires assignment of a new syntactic function to the semantic unit

in question, and a corresponding change in the syntactic structure allowing the final position of that element (e.g. the initial subject-agent of an active sentence becomes the final adverbial-agent in its passive transformation).

The two strategies of arranging semantic units into a linear sequence, i.e. the syntactic choice and the word-order choice, are clearly interrelated, since the final, or at least postverbal, position can be achieved by both, and a certain amount of word-order modification may occur within a given syntactic structure (e.g. fronting of a thematic adverbial). This is demonstrated by the existential-locative construction and subject postposition. Although neither is exclusively restricted to speaking or writing, they vary in frequency between the two modes or, to be more precise, in different registers of spoken and written language:

- (1) *There is a rusted car at the end of the row.*
- (2) *At the end of the row is a rusted car.*
- (3) *At the end of the row there is a rusted car.*

It is worth noting, however, that final subjects are common in spoken language with fronted adverbials such as *here* and *there*.

- (4) *Here comes the first question.*

5. The working hypothesis for this article was that spoken and written language would be different in the frequency of deviations from the grammatical word-order and possibly also in the placement of the rhematic element. The principle of *end-focus* was expected to operate more prominently in written than in spoken language, where the signalling role of position in the sentence can be overridden by prosodic means, namely by the position of the intonation nucleus.

The data for the analysis come from four different sources of authentic British English, two for written language (a book of fiction and three newspaper articles), and two for spoken language (extracts from two discussion programmes available as podcasts on the BBC Radio 4 webpage). Each source yielded 100 units for analysis; in written language the units usually coincided with sentences or clauses, in spoken language the delimitation was, predictably, somewhat more problematic. The relevant sections of the radio programmes were first transcribed and then segmented into units on the basis of a combination of structural clues (structural completeness) and prosodic signals (pauses, tone-unit boundaries). Structural factors of delimitation were taken into account for the purpose of comparison with written language – had only prosodic factors been used, the resulting units would have been very difficult to compare in terms of word-order with the relatively more regular structures retrieved from written language.

5.1. Neither of the samples contained a substantial proportion of deviations from the standard grammatical distribution of clause constituents. The most frequent of these included fronted adverbials and objects, which were slightly more common in writing. The results are presented in Table 1.

Table 1: Fronting

	Written		Spoken	
	Fiction	News	Museum talk	Gardening
Fronted A				
Time	5	2	2	1
Place	1	1	2	0
Other	0	1	0	1
Fronted O	2	0	1	0
Sub-total	8	4	5	2
Total	12		7	

Finite initial adverbial clauses were not included into the count, because they occur in the initial position naturally. On the other hand, both of the fronted objects in fiction were clausal, complementing a verb of speaking as indirect speech.

(5) *He might have been a bit tense, she had suggested, kindly, ...* (Bryers, 1987, 80)

Admittedly, their status as objects is not quite clear, since they may equally be interpreted as sentences with an inserted comment clause, clearly reminiscent of a pattern commonly found in direct speech followed by its introductory clause. The latter interpretation is corroborated by the absence of conjunction and the presence of a comma. Whatever the interpretation, these structures are characteristic of written language reporting instances of spoken language, such as in fiction or news reporting.

5.2. The uncertainty concerning classification of structures is by no means restricted to the above example, and definitely not to written language. In spoken language intonation is expected to serve as a means of disambiguation, but still there are a number of instances of mismatch between the structural indicators and the prosodic clues. This is not to say that this mismatch renders the structures incomprehensible, but it definitely raises doubts about how to interpret them as part of the language system. For example, it is by no means clear whether to interpret the following as an instance of fronting of the pronoun *this* or a repetition of subject (or at least part of it) with an intervening comment clause:

- (6) *You can now get the signature of jade so specifically that you know not only which region it came from but exactly where in the region. And this we now know – this axe – comes from a particular boulder on the top of a hill in northern Italy.* (Kendall, 2010, 3:07 – 3:22)

Both occurrences of the pronoun are strongly accented, with *we now know* being in the prosodic shade of the preceding pronoun, but not separated from it with a distinct pause. On the other hand, *this axe* is both preceded and followed by a distinct pause, as if the whole syntactic structure was broken into separate building blocks that were presented to the listeners to work out the structure and hence the intended meaning on their own. Related to the uncertainty concerning the syntactic structure is the question of reference of the first of the pronouns *this*; it can refer situationally to *axe*, or cataphorically to *this axe comes from a particular boulder...*, in which case it would be treated as an instance of rhematic fronting. Significantly, a conclusive solution remains unavailable and probably could not be provided even by the speaker himself.

The same extract contains a potential *wh*-cleft with an inserted comment clause.

- (7) *Well, what this object tells you is – and it's certainly very remarkable how people use and make things because this beautiful to hold (sic) I'm afraid I'm the only one allowed to touch.* (Kendall, 2010, 1:57 – 2:07)

The comment clause is underlined in the example, and the structure without it would read as follows:

- (8) *Well, what this object tells you is how people use and make things...*

However, since there is no prosodic break after *remarkable*, it can equally be interpreted as abandoning the *wh*-cleft structure unfinished and following with an extraposed subject clause introduced by anticipatory *it*. The latter interpretation is supported by the fact that additional information is presented in chunks which demonstrate little structural or even semantic link to the preceding text and in one instance a copular verb is missing: *this (is) beautiful to hold*.

The spoken extract about gardening includes an instance of potential left-dislocation of the subject *things like carrots and parsnips*, or, alternatively, an instance of apposition, where the phrase *things like carrots and parsnips* specifies what is meant by *roots*.

- (9) *The only thing I would say is – definitely do not dig it or incorporate manure the year before – you put your roots in – things like carrots and parsnips – if they come across a broken soil with bits of manure in it – then they tend to fork and make funny shapes.* (Robson, E., 2010, 8:33 – 8:47)

5.3. Another point of comparison was the functional sentence perspective of spoken and written discourse, respectively. The comparison consisted in establishing the position of the information focus, or the rheme in Firbasian terms, and the clause constituent carrying it, with regard to its position. In considering the rhematic position of clause constituents, it is important to bear in mind that all the clause constituents with the exception of the subject (and the verb itself) are normally postverbal. Where they are the only postverbal complements of the verb, they tend to be rhematic as well as final; when they occur as rhemes in their normal postverbal position and are followed by a non-rhematic constituent, they are treated as non-final. The subject behaves differently because its regular position is preverbal (marked here as non-final). Preverbal subjects are usually thematic and may be preceded by another thematic element, although rhematic preverbal subjects also occurred in the corpus.

- (10) *Every fourth year potatoes come round, every other fourth year roots, carrots and parsnips and so on, come round,...* (Robson, E., 2010, 8:33 – 8:47)

The final position of the subject is marked with an asterisk, suggesting that the subject occupies a postverbal rather than absolutely final position, when it follows the verb in the *there is* construction. In these positions, the subject tends to be rhematic even when it is followed by another constituent, typically an adverbial. No instances of final rhematic subjects in postposition were found in the corpus. The findings are presented in Table 2.

Table 2: Position of rheme and the rhematic constituents

	Written				Spoken			
	Fiction		News		Museum talk		Gardening	
	final	non-fin.	final	non-fin.	final	non-fin.	final	non-fin.
A	31	1	33	4	21	10	32	1
O	26	1	35	5	13	7	13	8
Cs	17	2	3	0	17	7	3	4
Co	4	0	2	0	0	0	0	0
V	5	10	9	5	11	8	4	22
S	3*	0	0*	4	2*	4	7*	6
Total	86	14	82	18	64	36	59	41

The differences between the two varieties seem significant enough to support the hypothesis that spoken language does not adhere to the principle of *end-focus* as strictly as written language because the three indicators of rheme in written language

(the contextual factor, the semantic factor and linear modification) are complemented by intonation; in other words, the signalling function of linear arrangement recedes into the background, making room for prosody to assert itself more prominently.

The absence of intonation in written language may occasionally be compensated for by graphical means, e.g. by the use of italics indicating a stress on a particular element (in the following example the highlighted item is underlined).

(11) *There were women like that.* (Bryers, 1987, 81)

Highlighting a particular element by graphical devices in writing has the same effect as the re-evaluative use of prosody in speaking (Firbas, 1999, 159); in the above example it assigns the rheme to an element that would not be interpreted as rhematic on the basis of the three non-prosodic factors of FSP.

6. According to Firbas, prosody in spoken language can override all the other factors. However, he is quick to point out that, even in such cases, the resulting interpretation of FSP remains the outcome of an interplay of all factors involved. In this respect, prosody cannot be judged on its own, but only in relation to the non-prosodic factors (Firbas, 1999, 148), confirming an FSP structure suggested by the non-prosodic factors, resolving instances of potentiality, or re-evaluating the interpretation.

In strictly spoken discourse, prosody is affected by lack of planning in a similar manner as syntactic structure is. While in reading a piece of text aloud, prosody is superimposed onto what is essentially written discourse, in spontaneous conversation structure and prosody are forged at the same time, but not necessarily identically. The units are shorter, and a single structural unit may be divided into two or more prosodic units, reflecting the step-by-step process of constructing structure. Natural spoken language may therefore convey the impression of a succession of short, loosely connected sequences, delimited by prosodic rather than structural boundaries, as if the speaker felt the need to complete the units prosodically even when they are not yet complete structurally. This may occasionally result in informationally weak elements receiving prosodic prominence and consequently in the loss of the rheme-indicating function of prosody. The rheme must then be identified on the basis of the interplay of non-prosodic factors. The following extract shows multiple instances of extremely strongly stressed verbs (underlined), most of which are clearly non-rhematic; other intonation nuclei coinciding with actual rhemes are not marked in the extract to highlight the unusually high proportion of prosodically prominent verbs. Interestingly, many of the rhematic peaks are prosodically less distinct than those carried by the non-rhematic verbs.

- (12) *It's a wonderfully strokable, holdable object. Erm, and it's meant to be an axe with which you chop wood, trees This is the kind of stone axe with which the people who came, the farmers that came to Britain about six thousands years ago, chopped down trees, made a wooden world and made the world that we now know. This is what created the green and pleasant land. Erm, but this one was never used. This is an object that's been made to make you think about what an axe is rather than to use, completely intact, it's beautifully polished, shaped, very difficult to make, and it's there to be an object of contemplation, of desire above all.* (Kendall, 2010, 2:09 – 2:50)

The prosodic prominence of the verbs in the extract above cannot be explained by FSP factors. However, when the actual rhemes are taken into account, the prosodic peaks (rhematic and non-rhematic) show a remarkably regular overall distribution, at least in some sections, giving the speech a distinctly rhythmical quality. This suggests that the rheme-indicating function of prosody may occasionally recede into the background and be replaced by other functions, in which case the regular distribution of intonation nuclei is employed to create a strong dramatic effect. Without further research, it remains a matter of speculation whether such use of prosody is the property of an individual speaker's style, a feature of the register in question, or whether it is a natural phenomenon of spoken discourse.

7. Although straightforward comparison of word-order and FSP in speaking and writing is made difficult by the relative irregularity of structure in spoken discourse and by the weakness of border signals between units, the analysis did reveal differences between written and spoken language both in the frequency of deviations from the grammatical word-order and in the position of the rheme in the sentence, largely in accordance with the original hypothesis.

Written language, more than spoken language, tends to employ variation in the word-order to achieve a linear distribution of communicative dynamism, i.e. the *end-focus*. This is due to the absence of prosodic clues and, possibly, also due to the relatively greater complexity of structure. With more clause constituents complementing the verb in written language, there is more competition for the individual positions in the sentence, as well as a greater number of alternative arrangements available. On the other hand, spoken language, with fewer complements has less potential as well as less need for such variation. The presence of prosodic clues along with the relative simplicity and brevity of structure reduces the need for alternative clause element distribution.

Closely related to variation in word-order is the preferred position of the rheme. As stated above, written language favours the *end-focus* more strongly than spoken

language, employing word-order as an important means of achieving it, whereas spoken language shows a higher proportion of non-final rhemes, indicated, in addition to the three non-prosodic FSP factors, by prosodic prominence as an exclusive rheme-indicating device of spoken discourse.

A somewhat surprising observation concerns the role of prosody. While remaining a powerful indicator of the rheme in spoken language, at least under ideal circumstances, prosody was found to be affected by the structural irregularity of spontaneous spoken language. In some cases the particular distribution of prosodic clues seemed to result from the manner in which spoken language is produced: as a loosely connected sequence of relatively short and possibly structurally incomplete units, as if the speaker felt the urge to complete the units prosodically at least, in order to compensate for their structural incompleteness. As a result, with limited planning ahead, the speaker occasionally produces prosodic patterns that not only contribute little to the construal of meaning, in particular to the interpretation of FSP roles, but sometimes may even prove misleading.

Admittedly, the number of writers and speakers and the extent of the language samples explored were too limited to provide a safe ground for conclusive findings. Yet it is hoped that the differences revealed are not accidental and might inspire further, more detailed research. Such research promising potentially interesting results could focus on comparison of authentic spoken texts with written texts read aloud.

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Vladislav Smolka: WORD-ORDER VARIABILITY AND FSP IN WRITTEN AND SPOKEN DISCOURSE

The article explores the role of word-order as an indicator of the rheme in samples of written and spoken language respectively. The assumption was that this role of word-order would be more prominent in written language, where the choice of a particular linear arrangement may be prompted by the need to achieve the effect of end-focus. In spoken discourse the rheme is signalled by prosodic clues, namely by the placement of the intonation nucleus, irrespective of its position in the sentence, rendering word-order a secondary indicator. As a result, the frequency of deviations from the grammatical word-order, e.g. fronting, is lower in spoken discourse. However, the analysis of word-order in spoken discourse is made difficult by irregularity and structural incompleteness.

Key words: spoken and written discourse, word-order, FSP, prosody